LEONARD THOMAS AARONS-DITSON

AI ENGINEER

Applied AI · Agent systems · Data science & experimentation · Evaluation · Python / ML
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PROFESSIONAL SUMMARY

AI engineer and data scientist with a UCL physics and quantum-information research background, commercial machine-learning experience and a track record of building agent systems and testing whether they can be trusted. Python since 2013; TensorFlow/Keras from 2022. Recent work: a 162-episode tool-use evaluation whose trace audits exposed hidden model assistance in **31/48** development calls; a Poisson gradient-boosted forecaster that cut MAE **24%** against baseline on 13,440 predictions; a coding-agent review workflow that caught a defect missed by **four passing tests**; and a prize-winning hackathon build now in a VC accelerator.

CORE SKILLS

Data science & statistics: experiment design, A/B and quasi-experimental comparison, n-of-1 crossover designs, baselines and holdouts, leakage testing, calibration and uncertainty, Poisson regression, gradient-boosted trees, time-series forecasting, Bayesian updating, Monte Carlo simulation, pharmacokinetic/pharmacodynamic (PK/PD) modelling.

AI / ML: TensorFlow, Keras, scikit-learn, reinforcement learning, fine-tuning, LLM APIs (Claude, OpenAI), prompt and context engineering, RAG and retrieval, tool calling, MCP servers, multi-agent coordination, LLM-as-a-Judge, evaluation datasets and scoring rubrics, Claude Code and Codex CLI.

Engineering: Python, pandas, NumPy, SciPy, SQL (PostgreSQL, SQLite, DuckDB), FastAPI, Pydantic/JSON Schema, pytest, Git/worktrees, GitHub Actions, REST APIs, OAuth 2.0, Streamlit, TypeScript/React, Excel/CSV data pipelines.

Scientific computing: C/C++, CUDA, PyOpenCL, GPU/HPC, Qiskit, QuTiP, numerical optimisation.

EXPERIENCE

Hackathon builds & independent AI engineering | 2026

- **Reinvent the Wheel hackathon**, London, Oct 2026 — built an Isabelle/HOL VS Code extension and MCP server so a coding agent writes proofs the Isabelle kernel checks. Implemented the full extension and server solo; concept with Alex Hyman.
- **Cantor8 'Build on Canton'**, London, Aug 2026 — team of five; designed the mathematics and implemented the core of a collateral optimiser. **$500 prize** from a Canton-ecosystem judging panel; the team was admitted to Cantor8's VC accelerator (1-1 investor meetings ongoing).
- **Rebuild Private Markets (Ylookup × Encode)**, Sep 2026 — team build of an evidence-linked financial review tool: source-linked PDF/Excel checks, Decimal reconciliation, discrepancy/cannot-verify states and human approval.
- **Daytona HackSprint (Entrepreneurs First)**, Aug 2026 — team build of an RL agent teaching a spacecraft to avoid objects; diagnosed a zero-reward failure and fixed it with a braking reward term.

Selected projects (2026) — building agent systems and testing their reliability

- **Agentic Physics Bench** — 162-episode LLM tool-use benchmark across no-tool, optional-tool and required-tool conditions with frozen datasets, deterministic grading and per-call traces. Trace audits exposed undeclared second-model assistance in 31/48 development calls; excluded affected runs, added baseline checks and measured tool choice separately from accuracy.
- **FleetCast** — demand forecaster on NYC taxi data (20 Manhattan zones, 30-minute bins): DuckDB/scikit-learn pipeline, past-only features, leakage tests, chronological holdout. Poisson gradient-boosted trees: MAE 10.83 vs 14.29 persistence and 16.22 weekly-naive over 13,440 predictions; residual bias of ~+2 pickups traced to weekly-lag staleness. Streamlit error-analysis interface; 28 tests.
- **Coding-Agent Cross-Review** — harness comparing Codex and Claude Code on the same task with isolated Git worktrees, adversarial reviews and fresh acceptance checks (~150 tests). Cross-review caught a Unicode defect missed by four passing tests.
- **Grounded Agent Planning** — plan/validate/replan pipeline with strict JSON Schema, evidence checks and persistent records; independent red/blue LLM evaluation and human approval. Validators rejected unsupported claims in 151/251 first-pass plans across 259 runs on 257 real listings.
- **Agent Context Router** — retrieval and memory for fresh agent sessions: bounded context packets with source hashes, atomic revision-checked writes; offline routing policy benchmarked against BM25.

999 Emergency Call Handler | East of England Ambulance Service NHS Trust | Jan 2026 – Jun 2026

Live emergency call handling: structured triage, accurate information capture under time pressure, prioritisation and escalation across multiple systems.

Independent quantitative modelling (self-directed) | 2022 – 2025

- Built pharmacokinetic/pharmacodynamic models of a multi-drug regimen, fitted to longitudinal self-collected data, to time dosing against measured response.
- Modelled radiation dose gradients from treatment geometry to anticipate the timing and location of side effects.
- Ran an n-of-1 crossover across drug changes with washout periods and a scored side-effect log (Excel/CSV, pandas, TensorFlow/Keras); documented what single-subject data can and cannot support versus a randomised comparison.

- Medical Laboratory Assistant, Imperial College Healthcare NHS Trust (2023–24): sample handling, analytical and regulatory work in a hospital laboratory.
- Medical leave during this period; returned to full-time work in 2026.

Data Scientist | Sticky, London | Oct 2020 – Feb 2021

- Built reinforcement-learning, predictive and inference models in Python across customer, payment and product-placement data; modelling, analysis and automated decision tools contributed to a **27%** increase in profitability.
- Investigated SaaS incidents with product and operations teams, separating customer symptoms from data, software and process failures; communicated recommendations to non-technical stakeholders and in investor-facing discussions.

Postgraduate Quantum-Information Research | UCL | 2018 – 2021

- Spin-chain communication, density-matrix evolution and time-dependent open systems in Professor Sougato Bose's group; Python/NumPy and Qiskit simulations, parameter sweeps, custom quantum gates, Leggett–Garg inequalities.
- Quantum error-correction and qubit manipulation in Python with ML models for noise reduction and error detection; CUDA/PyOpenCL GPU kernels for numerical and linear-algebra workloads.

Earlier roles: Data Analyst, BPL Global (2017); project/data support, Canary Wharf Contractors (2016, Employee of the Month).

EDUCATION

University College London — PGCert Physics with Distinction, 2021; BSc Physics, 2018. Postgraduate study within the MSc Physics programme, 2018–2021: Advanced Quantum Theory 93.5%; Modelling Quantum Many-Body Systems 86%; assessed coursework in high-performance computing and machine learning 80%. Selected for the Cumberland Lodge academic programme (one of 15 UCL Physics students).

COMMUNICATION & DISTINCTIONS

Delivered lectures at Imperial and UCL, including quantum simulation for students, alumni and staff; quantum-computing article commended for clarity. Imperial College summer programme at 15–16: presented on emergence and contributed to a condensed-matter poster. Self-taught A-level Mathematics, Further Mathematics and Physics.

Competitive UCL rower: Cambridge Head-winning crew; second nationally in the novice category at Newcastle. Organised a 72-hour rowathon raising £25,000 for Cancer Research UK.